

Faithful Indoor-Localisation Benchmarks for Majid’s Four Papers

Early evidence release

Executable reconstruction and audit

7 August 2026

Abstract

This meeting-ready release contains completed reconstructed evaluations for MPUrge–MAP and SMART–LEA, paper-extracted protocols for all four papers, readable method recipes, the valid causal-HMM result, and the causal multi-level-map experiment. It clearly separates final reconstructed evidence, paper-reported context, and smoke diagnostics. The full release will add the completed original-MPUrge aggregation, corrected CNN rerun, richer per-condition tables, and closed source-provenance manifests.

1 Fairness rule: match acquisitions, not feature shape

This report does *not* impose a delay-only rule. A challenger may use delay, noisy received power, AoA/AoD, complex CIR/CSI, or another physically measurable feature from the same acquisition. Noise is applied coherently: lower SNR perturbs power, delay and angle and increases path dropout. We do not give a method clean power while corrupting only delay.

What is matched is the information and calibration budget. If Majid’s protocol gives N survey locations and Q query acquisitions, every challenger gets those same locations and acquisitions. It may not request a dense labelled map, thousands of ungranted simulator examples, unmeasured candidate fingerprints, exact simulator ray/VT identifiers, the true query coordinate, or a clean latent channel. A measurable transmitter channel or physical transmitter identity is allowed.

The original simulator projects, raw data and random seeds are unavailable. Thus our numbers are executable *replacement-scene evidence under a paper-shaped protocol*, not claims to exactly reproduce a published number. Published values are shown only as context and never pooled with reconstructed results.

2 Completed MPUrge–MAP leaderboard

The locked suite has 16 conditions, 100 held-out queries per condition and 25 methods: 40,000 primary rows plus 33,600 learned-seed rows. Every method sees the same 80 independently acquired survey fingerprints and queries. Mean error is

$$\text{ME} = Q^{-1} \sum_q \|\hat{\mathbf{x}}_q - \mathbf{x}_q\|_2.$$

Intervals use a paired clustered bootstrap over identical physical queries. Gain is corrected Majid-style error minus challenger error; positive is better.

Method	Mean (m)	Gain (m)	95% paired CI (m)
Beta-survival multimodal	0.2528	0.8110	[0.7224,0.9033]
Survival-CIR + ToA ExtraTrees	0.2601	0.8036	[0.7146,0.8963]
Analytic-anchor residual multimodal graph	0.2745	0.7893	[0.6973,0.8843]
Candidate self-attention, multimodal	0.3349	0.7289	[0.6452,0.8133]
Genuine candidate cross-attention, multimodal	0.3434	0.7204	[0.6373,0.8063]
Assigned-VT receiver consensus with AoA	0.3463	0.7174	[0.6088,0.8257]
PointNet multimodal direct regressor	0.4373	0.6264	[0.5220,0.7331]
Multimodal feature ExtraTrees	0.4480	0.6158	[0.5246,0.7076]
DICHASUS-style coherent ADP + ToA k NN	0.7746	0.2891	[0.1858,0.3923]
RRLE/map-aware mixture of experts	1.0445	0.0193	[−0.0137,0.0506]
Path-count special case only	1.0589	0.0049	[−0.0036,0.0133]
Corrected Majid-style MPUrge–MAP	1.0638	0	reference
Temperature softmax only	1.0721	−0.0084	[−0.0149,−0.0016]
Subset-consensus head	1.0752	−0.0115	crosses zero

Method	Mean	Gain	95% paired CI
PointNet analytic reranker, delay	1.0961	−0.0324	crosses zero
Symmetric Chamfer + inverse top-3 only	1.1077	−0.0440	crosses zero
Spatial graph diffusion, delay	1.1084	−0.0446	confirmed adverse
Analytic residual graph, delay	1.1252	−0.0614	adverse
Combined Chamfer/temperature/path-count	1.1345	−0.0708	adverse
EvoMDP feature/rank fusion	1.1353	−0.0715	adverse
Delay-only ExtraTrees	1.2039	−0.1401	adverse
MCA nearest fingerprint	1.3143	−0.2506	adverse
Original MPUrge P5, corrected window	1.3160	−0.2522	adverse
CAEZ-style delay probability MLP	1.3387	−0.2749	adverse
PointNet delay-only direct regressor	1.3509	−0.2872	adverse

The corrected conclusion differs from an earlier headline. Chamfer alone did *not* improve the corrected baseline here, so the old 0.4992 versus 1.4138 m claim is retracted as a general component claim. The robust positive result is the family that uses survival-aware multimodal information from the same acquisition. Behavioral leakage, ordering, empty-set, duplicate-path and finite-output audits pass. The run-recorded source hash is retained; a shared receiver file was edited after launch, so byte-for-byte closure against the later disk copy remains an explicitly open provenance item.

3 Completed SMART–LEA reconstruction

The full run contains 24 scenarios: seeds {29, 47, 71}, clean and matched-noisy acquisition, and survey spacings {0.25, 0.5, 1.5, 2.5} m. It evaluates 30 methods over 720,000 raw prediction rows. Integrity checks pass. The table reports the best eligible method at each spacing; RMSE is shown because a rare large tail can make it disagree strongly with the mean.

Channel	Spacing	Best reconstructed method	Mean	RMSE
clean	0.25	fixed-height 3-D geometric-cross LEA-4	0.0198	0.0224
clean	0.50	fixed-height 3-D geometric-cross LEA-4	0.0562	0.8452
clean	1.50	range-power–AoA assigned-VT consensus	0.3197	0.3715
clean	2.50	range-power–AoA assigned-VT consensus	0.8576	1.2751
matched noisy	0.25	delay+CIR coherent method	0.1338	0.1524
matched noisy	0.50	multimodal query–reference attention	0.2266	0.3131
matched noisy	1.50	range-power–AoA assigned-VT consensus	0.4931	0.5753
matched noisy	2.50	range-power–AoA assigned-VT consensus	1.0460	1.9119

The large 0.50 m clean RMSE despite 0.0562 m mean reflects a very rare catastrophic tail (median 0.0187 m, 90th percentile 0.0342 m); it must not be hidden. In the same reconstruction, Majid-style LEA-4 means at spacings 0.25/0.5/1.5/2.5 are 0.0369/0.0791/6.2071/9.4343 m clean and 0.9224/1.8427/6.8121/9.7291 m noisy. MCA means are 0.1018/0.2380/6.3847/9.4690 m clean and 0.9929/1.9665/6.9322/9.6041 m noisy. These are comparisons inside the replacement 25 × 25 m scene, not against the paper’s exact floor plan.

For context only, the paper reports MCA/LEA-1/LEA-4 mean errors at 0.25 m spacing of 0.14/0.04/0.01 m, and at 2.5 m spacing of 1.73/1.45/1.30 m. The reconstructed trend is much harsher at sparse spacing, which is exactly why the scene boundary must be explicit.

4 MPUrge VT and CNN status

4.1 Original MPUrge virtual-transmitter mapping

A reflected path with range $r = c\tau$ and arrival direction $\mathbf{u}$ proposes a virtual transmitter $\mathbf{v} = \mathbf{x} + r\mathbf{u}$. MPUrge associates proposals across an ordered calibration sequence, retains persistent VTs and purges transient clutter. The corrected code uses total window $P = 5$ (half-window two), paper conflict ordering, per-transmitter support and dynamic threshold descent until exhaustion. Unknown cardinality is primary; true cardinality is diagnostic only.

The smoke run exercised all 13 executable branches and all 26 dynamic survival thresholds. Bernoulli/RFS tracking obtained **diagnostic** VT $F_1 = 0.9554$, symmetric-Chamfer set error 0.0559 m, and downstream localisation 0.286 m. These are branch diagnostics, not a final leaderboard; the full aggregation is audited separately. Exact ray/VT IDs are forbidden. A supervised VT PointNet requiring 5,000 extra labelled simulator examples is excluded because of extra information budget, not because PointNet or multimodal features are forbidden.

4.2 CNN paper

The paper-shaped training set has 20 reference points, 60 AP positions and seven obstruction configurations (8,400 labelled fingerprints per room), at 60 GHz, 2 GHz bandwidth, nominal 20 dB SNR and at most nine paths. The paper reports Room-A RMSE 1.82 m classifier/1.84 m regressor versus MCA 2.58–2.60 m; Room-B 2.14/1.47 m versus MCA 3.64–3.65 m. These are [paper-reported](#) values.

Our corrected design separates a privileged *paper-condition-map* bracket, which is told the true obstruction map, from a deployable *environment-blind* bracket, which pools all maps and selects only from observed noisy fingerprints. Validation maps are now fit-only, and validation queries use independently corrupted repeat acquisitions, preventing identity self-matching. Older smoke values are intentionally not ranked because this correction changes selection; the corrected rerun belongs in the full release.

5 Four protocols in plain language

SMART-LEA. Input is one independently noisy array-CIR per measured survey/query location and transmitter; delay, power and AoA are extracted from it. A sparse reference map is built, locked off-grid positions are test-only, and the output is a receiver coordinate. Report mean, median, RMSE and tail percentiles by spacing/noise. LEA constructs one local candidate per transmitter, ranks them by residual, then combines the best L ; LEA-4 uses four.

MPUrge. Input is an ordered sequence of path sets plus known *calibration* receiver poses. Output is a persistent VT set and downstream location. Held-out query positions are never used for association. Score set precision/recall/ F_1 , Chamfer set error, map cardinality and downstream position error.

MPUrge-MAP. Input is a sparse map of 80 measured reference fingerprints and one noisy query. References and locked off-grid queries are disjoint; learned choices use separate validation seeds/queries. The output is 2-D position and the primary score is paired Euclidean error. Empty/missed/duplicate path cases have fixed finite behavior.

CNN. Input is a fixed-length tensor of up to nine observed paths for an AP-receiver link. Training uses the paper-shaped $20 \times 60 \times 7$ links per room; test uses 1,600 spatially off-grid links. The network outputs either one of 20 reference classes or continuous (x, y) . Report RMSE and the complete Euclidean-error distribution, separately for privileged and environment-blind brackets.

6 Reproducible explanations of the method families

MCA and corrected MPUrge-MAP. MCA returns the coordinate of the measured reference with smallest fingerprint distance. MPUrge-MAP keeps the three smallest-score references and computes

$$\hat{\mathbf{x}} = \frac{\sum_{i \in \mathcal{N}_3} (s_i + \epsilon)^{-1} \mathbf{x}_i}{\sum_{i \in \mathcal{N}_3} (s_i + \epsilon)^{-1}}.$$

The score, tie behavior, padding, empty sets and ϵ must be frozen before test.

Symmetric Chamfer and subset heads. For path sets A, B ,

$$d_{\text{Ch}} = \frac{1}{2|A|} \sum_{a \in A} \min_{b \in B} \rho(a, b) + \frac{1}{2|B|} \sum_{b \in B} \min_{a \in A} \rho(a, b).$$

The two directions penalize missing and extra paths. Temperature weighting uses $w_i \propto e^{-s_i/T}$ with validation-frozen T . Subset consensus forms estimates from multiple top-candidate subsets and chooses the most internally consistent. None improved the corrected MAP baseline here.

Spatial graph and analytic residual. Connect measured survey positions to spatial neighbors, with Laplacian L . Smooth evidence by $(I + \lambda L)\tilde{z} = z$. A safer learned version starts from analytic $\mathbf{x}_0$ and predicts only $\Delta\mathbf{x}$, so $\hat{\mathbf{x}} = \mathbf{x}_0 + \Delta\mathbf{x}$. Multimodal residual learning ranked third; delay-only graph smoothing oversmoothed scene boundaries.

PointNet/DeepSets. Embed paths independently and pool:

$$h = \text{pool}_j \phi(p_j), \quad \hat{\mathbf{x}} = g(h).$$

Max/mean pooling makes h invariant to path order. A direct head predicts coordinates. A candidate reranker combines query/reference embeddings, their difference and the measured reference coordinate, then predicts compatibility. Multimodal direct PointNet was strong; delay-only direct PointNet failed.

Genuine cross-attention. Query paths form Q , candidate-reference paths form K, V :

$$\text{Attn}(Q, K, V) = \text{softmax} \left(QK^\top / \sqrt{d} + M \right) V,$$

where M masks padding. Pool attended query tokens to one score per *measured* candidate, then softmax scores into coordinate weights. Random path permutations in training and evaluation verify invariance. The 0.3434 m result is real attention and does not inspect unmeasured candidates.

Candidate self-attention. Each measured candidate is a token containing its coordinate, analytic score and query-reference features. A Transformer lets candidates compare before producing weights. No candidate index encoding is supplied, and evaluation permutes candidates. It scored 0.3349 m.

Beta survival and Bernoulli/RFS. A path track has survival $\pi \sim \text{Beta}(a, b)$. Detection increments a ; a credible miss increments b . Marginalizing π avoids deleting a reflector after one dropout. In the RFS version, every VT also has existence probability r , updated from detection, clutter and survival likelihoods; birth/confirmation/deletion are probabilistic. Association uses observable geometry/features, never true ray identity.

DiffAssign-K. Build an observed-path-to-map cost matrix, add dummy rows/columns for birth/miss, and use Sinkhorn normalization for a soft nearly doubly-stochastic assignment. Choose cardinality K by a frozen BIC-like penalty; true cardinality is never primary.

Assigned-VT consensus. Every assigned VT/path gives a receiver hypothesis, e.g. $\mathbf{x}_j = \mathbf{v}_j - c\tau_j \mathbf{u}_j$. Robust inverse-residual weights fuse the hypotheses. This analytic use of same-acquisition AoA reached 0.3463 m.

Coherent ADP and Survival-CIR. Coherent ADP compares normalized complex angle-delay profiles made directly from noisy array CIRs, then interpolates the nearest measured coordinates. Survival-CIR feeds fixed summaries of CIR coherence, delay, noisy power, angle and track survival into ExtraTrees. Hyperparameters are validation-frozen. Its 0.2601 m performance, close to Beta survival, points to survival-aware multimodal representation rather than one special neural architecture.

EvoMDP and RRLE. EvoMDP combines frozen distances/ranks using coefficients selected only on validation scenes. RRLE is a map-aware mixture of analytic, nearest-neighbor and learned experts whose router sees uncertainty and map-density features. Neither produced a confirmed MAP gain here.

7 Sequential supplement: causal analytic HMM

For ordered trajectories, combine the analytic observation likelihood with a causal motion prior:

$$p(\mathbf{x}_t \mid \mathbf{y}_{1:t}) \propto p(\mathbf{y}_t \mid \mathbf{x}_t) \sum_{\mathbf{x}_{t-1}} p(\mathbf{x}_t \mid \mathbf{x}_{t-1}) p(\mathbf{x}_{t-1} \mid \mathbf{y}_{1:t-1}).$$

The frozen HMM scored 0.3802 m versus 0.9654 m for framewise analytic MAP: gain 0.5852 m, 95% CI [0.4958, 0.6783] m, over 270 physical frames/1,350 condition rows. It uses neither future measurements nor query truth. It applies only when temporal ordering is available, so it is not smuggled into static leaderboards.

8 Multi-level/irregular maps: causal positive findings

The selection policy sees 72 possible sites but a site’s fingerprint is hidden until it buys that measurement. Thirty-six protected query fingerprints/coordinates never enter selection. Policies use nested acquisition paths and budgets 12, 20 and 30; the audit found zero reads of unselected survey features.

At budget 12, geometry-first farthest sampling plus finite-element interpolation was best: 1.5251 m; no adaptive policy improved it. At budget 20, causal FEM-variation selection plus polynomial/RBF localisation gave 1.4745 m

versus farthest 1.5463 m, a point gain 0.0718 m with group-bootstrap CI $[-0.1534, 0.3035]$ m. At budget 30, causal GP-variance plus polynomial/RBF gave 1.3434 m versus 1.4271 m, point gain 0.0837 m, CI $[-0.1002, 0.2674]$ m. The latter two are promising point estimates, not statistically confirmed wins.

The practical policy is: first acquire geometric coverage; only after enough measured sites support a stable field model, place later fingerprints at high causal interpolation variation/posterior variance. The earlier 1.5935 m “goal-surplus” claim is retracted because it inspected fingerprints at unmeasured candidate sites. Measuring a dense map and later discarding points is also not calibration saving.

9 Top three defensible ideas

1. **Survival-aware multimodal fingerprints:** Beta survival and Survival-CIR reduce reconstructed MAP error from 1.0638 m to 0.2528–0.2601 m, with paired intervals far from zero.
2. **Physics anchor plus learned correction:** the multimodal residual graph reaches 0.2745 m, learning a bounded correction instead of unconstrained coordinates.
3. **Permutation-invariant candidate reasoning:** self/cross-attention reaches 0.3349/0.3434 m using only measured candidates and survives order-invariance tests.

10 Release ledger

Suite	Status	Claim boundary
MPUrge-MAP	full results complete	40,000 primary rows; bootstrap and behavioral audits pass; source-copy hash closure pending.
SMART-LEA	full results complete	24 scenarios, 720,000 rows; replacement scene; recorded receiver hash differs from later edited disk copy.
MPUrge VT	full audit pending	Smoke table is diagnostic only.
CNN	corrected rerun pending	Published values included; leakage-affected old smoke excluded.
Causal HMM	complete	Sequential protocols only.
Irregular map	complete	Causal audit passes; no confirmed adaptive win.

Paper identifiers

M. Abdmoulah and E. Steinbach: *MPUrge*, DOI 10.1109/IWCMC65282.2025.11059597; *MPUrge-MAP*, DOI 10.1109/IWCMC69287.2026.11580087; *SMART-LEA*, DOI 10.1109/IWCMC69287.2026.11579893. M. Abdmoulah, N. Neji and E. Steinbach: CNN fingerprint localisation, DOI 10.1109/IWCMC69287.2026.11580049.